

Plotting a Line of Best Fit for the Mass-Luminosity Relationship of Main Sequence Stars

1. Introduction

Through this report, I outline the approach I took to interpret and analyze the mass-luminosity relationship of stars on the main sequence.

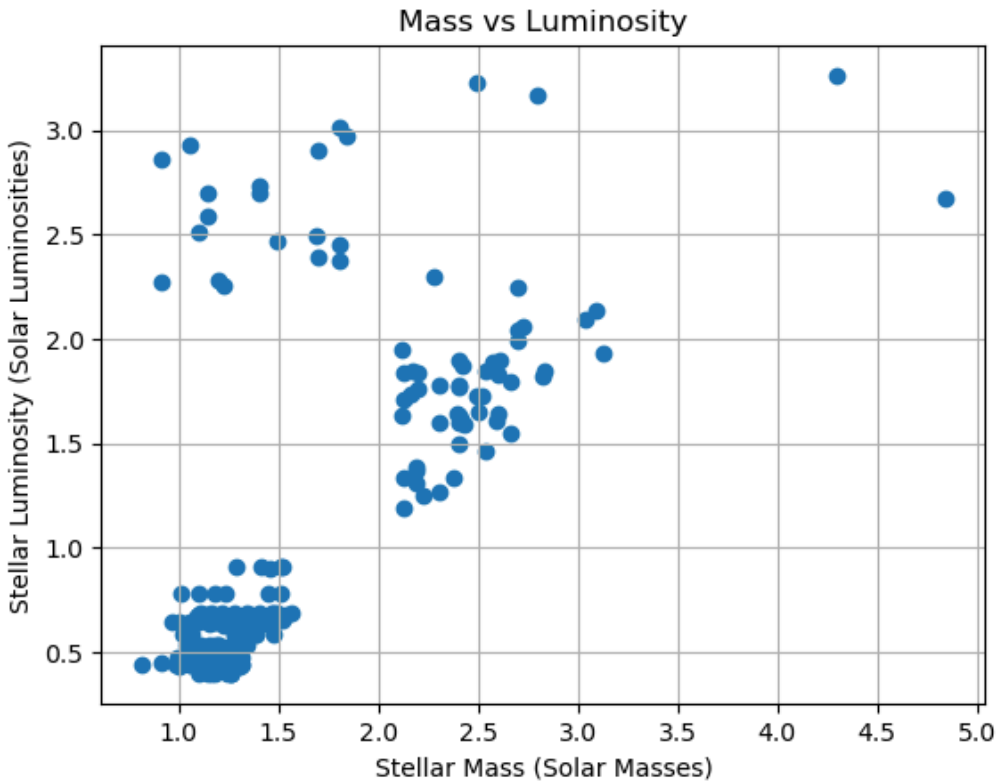
2. Chosen Phenomenon and Data Source

As seen from prior research and testing, the mass-luminosity relationship of stars on the main sequence follows a rough pattern that corresponds to this proportionality:

$$L \propto M^{3.5}$$

- L = Luminosity (in solar luminosities)
- M = Mass (in solar masses)

I pulled all data from the [NASA Exoplanet Archive](#) by uploading data from *all* stellar hosts. I originally intended to pull data from hosts that had data for both mass and luminosity, but this proved to be ineffective. The graphs I generated from these hand-picked data points were scattered and didn't seem to represent the full picture. Here's an example:



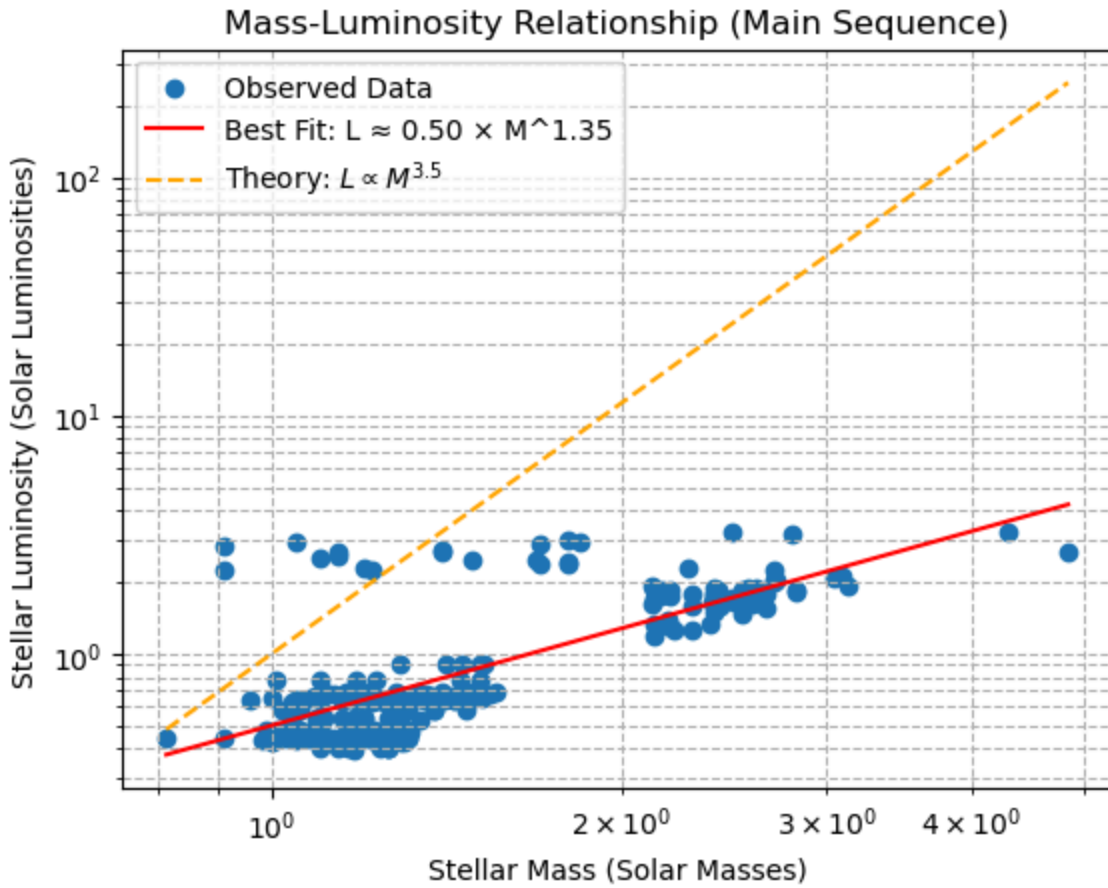
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3. Explanation of Model

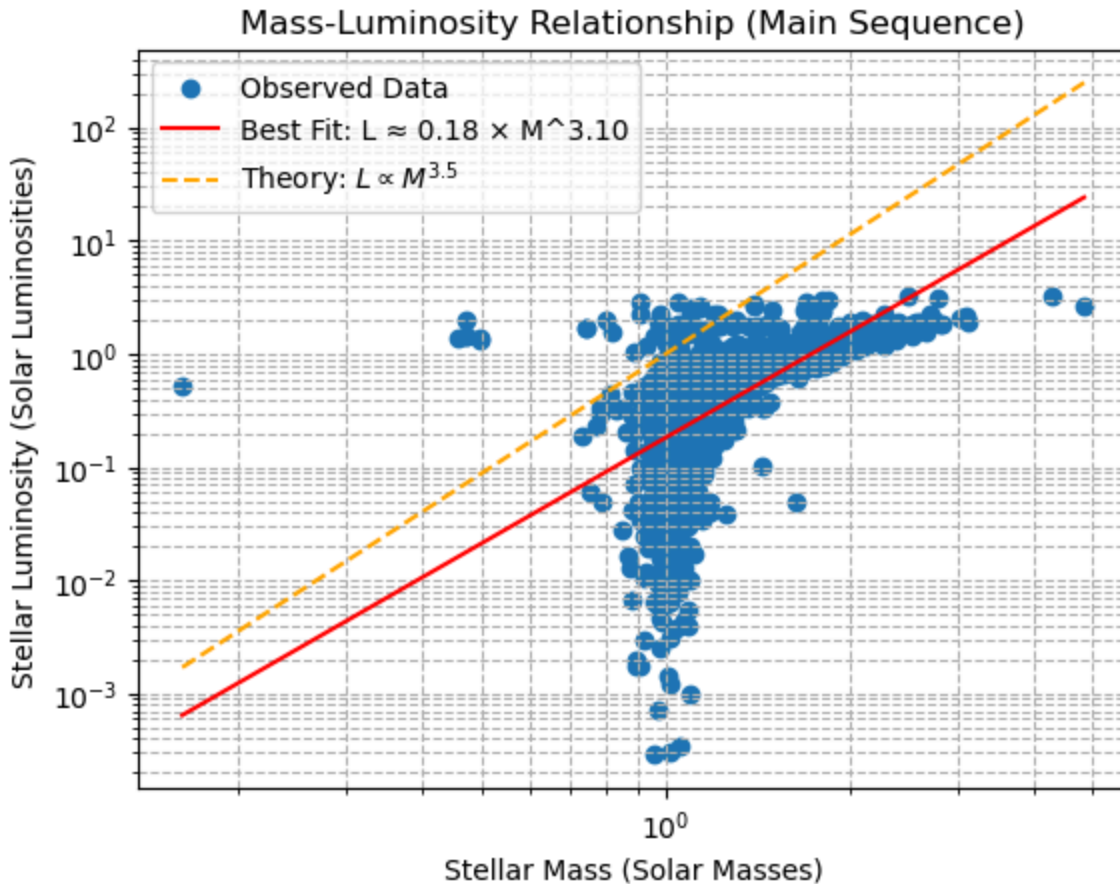
Pandas was used in the project to analyze the chosen data points, Matplotlib was used to plot those data points onto a simple graph (luminosity on the y-axis and mass on the x-axis). I will use the function `linregress` to look for a line of best fit (by providing an equation and data points).

Then, I compared the graph that I generated to the expected relationship ($L \propto M^{3.5}$). It first looked like this:

¹ As noted, this graph is overwhelmingly concentrated in the bottom left corner, which is not the result I was hoping for. This likely meant that I was somehow inadvertently skewing the data, or previous findings were wrong. I settled on the first option, and remedied this via importing more data points later on.



This looked off. I didn't understand why the disparity was so great between my theoretical line and the line I actually created. I thought it might be due to the possibly skewed data points, so I plotted again.



This time, I used all data points from NASA's exoplanet website. My reasoning for this was that if a stellar host only had luminosity data or only had mass data, then it wouldn't be plotted, so I could import all data, and only the points with *both* mass and luminosity data would be plotted. This was effective, and as can be seen above, generated a graph that had a line of best fit that was much more aligned with the theoretical equation than before.

My generated line emerged with an equation of:

$$L \approx 0.18 \times M^{3.10}$$

This is only slightly off from the predicted $L \propto M^{3.5}$, as this would be $L \propto M^{3.10}$